

Lifetime Asset Allocation in New Zealand: A Simulation and Optimization Approach

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1 Foreword and Research Context

This research is conducted in partnership with Consilium NZ LTD (hereafter, Consilium). Consilium is a New Zealand-based financial services company that provides tools and consulting services to financial advisers and their clients in areas including investment management, portfolio construction, risk management, and financial planning. Consilium also offers managed funds and portfolio solutions designed to support advisers in meeting a wide range of client needs. The aim of this research partnership is to examine and improve lifetime asset allocation strategies for individual investors, with particular focus on retirees in the New Zealand context. Our objective is to produce insights and recommendations that may help Consilium better support advisers and their clients.

Although this research is conducted in partnership with Consilium, the views expressed in this paper are ours and do not necessarily reflect the official policy or position of Consilium. We retain full academic independence in the design, analysis, and reporting of this work. We also retain full ownership of the intellectual property associated with this research, while granting Consilium a non-exclusive licence to use the findings internally and share them with clients and partners. No confidential or proprietary information provided by Consilium has been used in this study.

We thank Consilium for their support and collaboration in this research project, and for the feedback that has helped shape the direction and focus of this work.

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2 Introduction

Managing a retiree’s nest egg requires navigating two broad forms of financial risk. A well-established principle in investment research is that assets with higher expected returns tend to come with higher volatility (e.g., (Campbell, 1996; Ghysels et al., 2005; Lundblad, 2007)). Although many investors spend considerable energy selecting particular securities or funds, Ibbotson and Kaplan (2001) find that roughly 90% of the variation in fund returns is explained by the allocation to major asset classes such as equities, bonds, and cash. As a result, asset allocation is the primary driver of a portfolio’s long-term behaviour. For the typical investor, whose retirement savings are usually held in managed funds that do not allow meaningful security selection, the most important strategic decision is choosing the overall exposure to these broad asset classes.

The central challenge for individuals approaching retirement is finding an allocation between safe and risky assets that provides the returns required for a comfortable lifestyle, while avoiding excessive exposure to loss. Achieving this balance is not straightforward. Once the allocation is considered as a function of risk appetite, savings capacity, spending needs, lifespan, and total accumulated wealth, the problem becomes complex. Many of these inputs differ significantly between individuals, and even small differences in behaviour or circumstance can lead to materially different outcomes. The mathematical problem of finding an optimal allocation through time is therefore highly sensitive to real-world considerations. This complexity has contributed to the emergence of various heuristics and rules of thumb that attempt to provide simple guidance in the absence of a fully individualised solution.

The significance of this problem extends beyond individual welfare. The global asset-backed pension market was valued at 68.9 trillion USD in 2024 (OECD, 2025). For many people, their retirement savings represent the largest financial asset they will ever manage apart from their home. Consequently, the question of how to allocate this wealth over the course of a lifetime is significant not only for individuals but also for governments. Even modest improvements in the efficiency of retirement strategies can reduce pressure on public pension systems. In New Zealand, for example, every resident aged 65 or older receives New Zealand Superannuation regardless of income or assets. The programme is funded through general taxation. Due to an ageing population and ongoing increases in life expectancy, the long-term fiscal sustainability of this system is becoming increasingly challenging. A recent Treasury report (The Treasury, 2025) estimates that spending on New Zealand Superannuation and healthcare could account for nearly half of all government expenditure by 2065 if current policy settings remain unchanged. This has led to suggestions that New Zealanders will need to save more for retirement, for example through higher KiwiSaver contributions, and that adjustments to eligibility age or means testing may be required. There is already a broad consensus that the public is under-saving for retirement (e.g., (Dexter, 2025; Gilbert, 2025; Luxon, 2025; Palmer, 2025)).

A critical requirement for the success of such policies is ensuring that individuals make appropriate decisions about their lifetime asset allocation. At present, the New Zealand government provides little guidance on how individuals should allocate their retirement savings. The Financial Markets Authority requires that default KiwiSaver funds adopt a Balanced risk profile (Financial Markets Authority, 2023), defined as holding between 35 percent and 63 percent growth assets (Financial Markets Authority, 2024). This replaced the previous Conservative default in 2021 (Retirement Commission Te Ara Ahunga Ora, n.d.), after it became clear that many KiwiSaver members did not switch funds from the default fund and that a conservative allocation was poorly aligned with the needs of long-term investors (Ministry of Business, Innovation and Employment and The Treasury, 2019). Although a Balanced fund is a reasonable one-size-fits-most solution, it is likely too conservative for many younger investors who have long investment horizons and the capacity to tolerate short-term volatility.

This paper has three objectives. First, we review the current literature on lifetime asset allocation, with particular attention to the glide-path strategies that dominate both academic discussions and industry practice. We also examine the different risk measures and optimisation frameworks that researchers have used to evaluate such strategies. Second, we analyse the spending needs and risk concerns of typical retirees, with a particular focus on the New Zealand context. Understanding these patterns is essential for designing realistic simulations of retirement behaviour and for evaluating the amount of risk that individuals can tolerate. Finally, we develop

and test a flexible simulation and optimisation framework for lifetime asset allocation. Building on the work of Mäkinen and Toivanen (2024), who propose a Monte Carlo portfolio optimisation framework, we adapt and extend their method to address the specific challenges that arise in lifetime allocation problems. We also modify their risk measure so that it reflects the practical concerns of real-world investors.

Our aim is to show that this framework can generate stable and robust strategies, while remaining flexible enough to incorporate a variety of risk measures, asset return models, and spending assumptions. In doing so, we aim both to test the validity of existing glide-path recommendations and to provide a foundation for further research in this area of personal finance.

2.1 The Status Quo on Lifetime Asset Allocation

In order to understand the rationale for any allocation strategy, we first require a clear picture of the financial life-cycle faced by a typical investor. Broadly, this can be divided into two phases: accumulation and decumulation. During accumulation, which typically spans early adulthood to retirement, investors build wealth through regular contributions and investment returns. Decumulation begins at retirement and continues until death; during this phase, investors withdraw funds to finance living expenses and other needs. This framing implies a life-cycle wealth trajectory in which wealth accumulates during working life, peaks around retirement, and may decline thereafter. Although this trajectory is often assumed to be hump-shaped, the extent to which real-world investor wealth actually declines in retirement (or whether it typically declines at all) remains an open question, to which we return later.

Another important concept in this discussion is human capital, which refers to the present value of an individual’s future earnings potential. During the accumulation phase, human capital is typically high, as the investor has many working years ahead. As the investor approaches retirement, human capital declines, and at retirement, it effectively becomes zero. This shift in human capital has important implications for asset allocation decisions, as it affects the investor’s overall risk profile and capacity to absorb investment losses.

The sheer amount of complexity in identifying optimal life-cycle allocation strategies is substantial. The optimal strategy depends on many factors and, likely because of this complexity, a large number of rules of thumb and financial products have emerged to guide investors and advisers. Most of these rules recommend a “declining-risk” strategy, in which exposure to equities decreases with age. This idea has been influential for decades and was even considered as part of potential reforms to the United States Social Security system under the George W. Bush administration (R. Shiller, 2005; R. J. Shiller, 2006). The fact that it was considered for such a large and consequential public pension system highlights the appeal, to both policymakers and governments, of improving retirement outcomes through better asset allocation.

A well-known example of this type of recommendation is the family of “age in bonds” rules, which suggest holding a bond allocation equal (or proportional) to the investor’s age. The implied life-cycle policy is approximately linearly declining in risk exposure: high equity weight at younger ages, decreasing over time, hence “declining-risk” or “declining-equity”. This approach appears in investment textbooks (e.g., (Bengen, 1996; Bodie et al., 2023)) and in popular advice literature (Choi, 2022). Dolvin et al. (2010) review U.S. target-date funds and find that most follow some form of declining-risk structure, often resembling a 120-minus-age equity rule. In New Zealand, SuperLife’s Age Steps product follows a similar pattern, reducing equity exposure from 99% at age step 18 to 10% at age step 80 (see Figure 2.1).

The intuitive logic behind declining-risk strategies is that older investors have less time to recover from losses, and that their spending needs become more immediate. In this view, reducing exposure to risky assets as retirement approaches is considered prudent. However, continuing to reduce risk exposure throughout retirement may not be optimal, and academic support for this approach is mixed at best; much of the evidence instead indicates that constant or even increasing-risk strategies may be more effective for retirees undergoing decumulation.

The first mathematically rigorous treatment, by Merton (1969), showed that under lifetime consumption-utility maximisation, the optimal allocation is constant through time and does not depend on age. However, that conclusion relies on an investor withdrawing a fixed proportion of wealth each period (rather than a fixed dollar amount) and on asset returns being independent and identically distributed (IID) through time. Even so, using empirical retirement data, R. Shiller (2005) and R. J. Shiller (2006) find that declining-risk strategies may be inferior to a constant 100 percent equity allocation for retirees making fixed-dollar withdrawals. Moreover, Estrada (2014, 2015) find that declining-risk glide-paths tend to underperform both constant-risk and increasing-risk strategies across a wide range of risk measures, asset classes, and investor types. Using bootstrapped historical return data, they show that declining-risk strategies produce lower terminal wealth without meaningful downside-risk improvement.

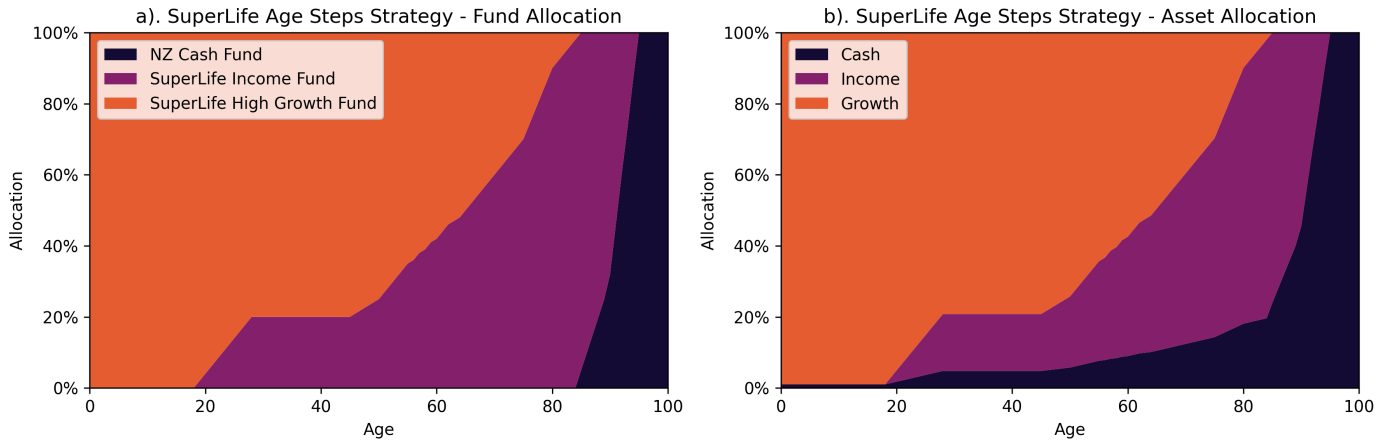


Figure 1: SuperLife's Age Steps target strategy as per their Statement of Investment Policy and Objectives (SIPO), showing the allocation to different funds (a) and the resulting allocation to different asset classes (b). Source: SuperLife (2026).

Although this paper will be principally concerned with the decumulation phase, it is worth noting that the evidence for declining-risk strategies during the accumulation phase is also mixed. The most robust theoretical support for declining-risk strategies during the accumulation phase comes from Bodie et al. (1992) and Cocco et al. (2005), who show that when human capital is taken into account, younger investors should hold a higher proportion of equities. In their models, human capital is treated as a bond-like asset because it provides a steady stream of income that is less volatile than equity returns. As investors age and their human capital declines, their overall risk exposure increases, which justifies a reduction in equity holdings. In this model, they essentially argue that the declining-risk strategy is appropriate during the accumulation phase, insofar as it offsets the changing risk profile due to declining human capital.

However, empirical evidence suggests that even during accumulation, declining-risk strategies may not be optimal. Anarkulova et al. (2025) find that for typical young investors with low wealth and savings rates, a constant 100 percent equity allocation provides similar downside protection to declining-risk strategies while delivering substantially higher expected returns. Using block-bootstrap resampled data from U.S. and international markets, they show that optimised glide-paths still underperform all-equity portfolios, except in the very earliest retirement years. They argue that fixed-income assets provide limited benefit once time-dependent return dynamics are incorporated, and that earlier studies tended to favour bonds because they assumed IID returns. In their optimal glide-paths, bonds appear only modestly, mostly at retirement onset, with allocations typically around 30–40 percent or less. This produces a U-shaped risk trajectory in which risky allocations dip at retirement and then rise.

This U-shaped risk trajectory is a common finding in the literature on optimal glide-paths, and it is often interpreted as evidence that some reduction in risk exposure at retirement is optimal. It appears again, more strongly, in work by Estrada (2015), who find that optimal glide-paths often involve a significant reduction in equity exposure at retirement, followed by a gradual increase in equity exposure later in retirement. This pattern is consistent with the idea that retirees are most vulnerable to sequence-of-returns risk at the start of retirement, when they have high accumulated wealth and low human capital. By holding some bonds at this point, retirees can reduce the likelihood of depleting their portfolio too quickly. However, as withdrawals reduce total wealth and the investor is further along in retirement, the impact of a run of bad returns becomes less severe (in real-dollar terms), and therefore a higher equity allocation becomes more acceptable in fixed-spending scenarios.

Although declining-risk strategies remain widely recommended in practice, the academic and mathematical literature does not strongly support their use during decumulation. Instead, the evidence generally suggests that constant or even increasing-risk strategies may produce better outcomes. While decreasing risk exposure may be appropriate near retirement during late accumulation, given heightened sequence-of-returns risk, the evidence suggests that once decumulation begins retirees may be better served by maintaining or even increasing risky-asset exposure.

The benefits of strategies such as SuperLife's Age Steps are therefore likely to be limited, and may even be counterproductive for many retirees. The broader evidence suggests that common guidance on life-cycle allocation may be incomplete, and that advisers may have scope to differentiate through more evidence-based

portfolio design. It should be noted, however, that the literature is not fully settled: conclusions still depend on the risk measure used, return-process assumptions, and the treatment of spending and withdrawals. Many existing models also remain limited in their representation of real-world investor behaviour and market dynamics.

Our goal will be to study and model investor concerns and behaviour in a way that is more realistic and relevant to the New Zealand context. We will then use this understanding to inform the design of a flexible simulation and optimisation framework for lifetime asset allocation, which can be used to test the validity of existing glide-path recommendations and to explore alternative strategies that may better meet the needs of New Zealand retirees.

Implications for Consilium NZ LTD

While Consilium does not currently offer a glide-path product, nor explicitly recommend a declining-risk strategy, the existing literature still has clear implications for their business. It suggests that prevailing views on life-cycle allocation may be incomplete, and that there may be scope for Consilium to differentiate through more sophisticated, evidence-based portfolio advice. In particular, there may be an opportunity to challenge adviser assumptions regarding the need for declining-risk strategies.

Consilium has typically based much of its modelling of portfolio performance on IID return assumptions, which may not accurately reflect real-world market dynamics. By investigating asset allocation strategies that account for more realistic return dynamics, Consilium can provide more robust and effective advice to their clients.

3 Retirement in New Zealand

Because the literature focused specifically on New Zealand retiree concerns remains limited, we also draw on international evidence. In doing so, it is important to outline the key economic and institutional features of the New Zealand retirement system, and to compare these with systems in countries from which we draw additional insights. New Zealand has a retirement system based on a universal public pension (New Zealand Superannuation) and voluntary private savings (KiwiSaver). The public pension provides a baseline income to all retirees, regardless of work history or asset position, and is adjusted annually for inflation. KiwiSaver, by contrast, is a voluntary savings scheme under which individuals can contribute a portion of their income to a retirement account, with additional contributions from employers and the government. Together, these mechanisms provide a basic safety net while encouraging personal saving and investment (Ministry of Social Development, 2022; Te Ara Ahunga Ora: Retirement Commission, 2021).

Given that KiwiSaver is relatively new, having been introduced in 2007, the profile of retirement concerns in New Zealand is likely to shift as more households retire with substantial KiwiSaver balances. Currently, 40 percent of New Zealanders aged 65 rely entirely on New Zealand Superannuation for retirement income, while a further 20 percent have only a small amount of additional savings (Retirement Commission Te Ara Ahunga Ora, 2022). As these balances grow over time, this demographic profile is likely to change. The government is also taking steps to increase KiwiSaver contributions, which may further increase retirement wealth for future cohorts (Luxon, 2025, e.g.). This likely implies a greater need for robust guidance on asset allocation and withdrawal decisions in retirement.

When drawing on international evidence, it is important to note that institutional settings differ across countries, and so external findings cannot be translated mechanically to New Zealand. Nevertheless, the growing role of KiwiSaver likely makes the New Zealand system increasingly comparable to Australia, which strengthens the relevance of Australian evidence for our purposes.

Australia has a retirement system based on a combination of a means-tested public pension and a mandatory private savings scheme (superannuation). However, the Australian system goes significantly further than New Zealand in encouraging private savings, with a mandatory contribution rate of 12 percent of income, compared to New Zealand's voluntary contribution system, and with a significantly higher contribution rate for those who do choose to contribute (Australian Taxation Office, n.d.; Swoboda, 2014).

3.1 Spending By Age in New Zealand

By far the best source of data on spending patterns in retirement in New Zealand is the Stats NZ Household Economic Survey (HES). Two reports have been published using this data that are particularly relevant for our purposes. The first is a report by Le and Richardson (2023) that provides a detailed analysis of expenditure patterns by age in New Zealand. The second is a more recent report by the Retirement Income Interest

Group (RIIG)(Retirement Income Interest Group, 2024) that focuses specifically on spending patterns through retirement and their implications for retirement planning and drawdown strategies.

Le and Richardson (2023) provide a detailed analysis of this pattern using New Zealand data. Using data from the Household Economic Survey (HES) for the year ended 30 June 2019, they find that expenditure falls around retirement and continues to decline thereafter. Mean total expenditure is \$64,700 for households aged 65–74, while means for households aged 75–84 and 85+ are 28% and 44% lower, respectively. Across categories, expenditure generally declines with age (with exceptions such as rent and household contents). The largest declines between younger and older retirees are in mortgage repayments, air travel, and recreation. Unusually relative to other countries, healthcare expenditure also declines with age in New Zealand, likely because much healthcare is publicly funded and therefore imposes lower out-of-pocket costs on most retirees. Overall, Le and Richardson (2023) find that older retirees spend significantly less than younger retirees, particularly in discretionary categories. These results are consistent with more recent HES-based evidence from Retirement Income Interest Group (2024), who estimate that retirement spending declines by roughly 2% per year in real terms after age 65.

What this data cannot tell us, however, is why this decline in spending occurs. There are essentially two possibilities. First, it could be that retirees are voluntarily reducing their spending because they become less active and therefore have less need for goods and services. The second possibility is that retirees are involuntarily reducing their spending because they become financially constrained. Without data that tracks individual retirees over time, it is difficult to disentangle these two effects. The HES data is cross-sectional, so it cannot track the same individuals as they age. Therefore, it is possible that the observed decline in spending with age is due to a combination of both voluntary reductions in spending and involuntary reductions due to financial constraints.

This decline in spending with age is important for our modelling of retiree behaviour, as it suggests that spending needs, and possibly risk tolerance, vary across retirement stages. Understanding why consumption falls, whether through voluntary lifestyle adjustment or binding financial constraints, is central to interpreting welfare implications. If the decline primarily reflects reduced welfare, then an optimal strategy might prioritise maintaining pre-retirement spending levels. If it mostly reflects voluntary adjustment, then forcing spending to remain on a pre-retirement path may be neither necessary nor desirable.

This decline in spending is a well-studied phenomenon in the economics of retirement and is sometimes known as the “retirement consumption puzzle”. This refers to the repeated empirical finding that expenditure often drops when people retire, despite standard life-cycle and permanent-income models predicting relatively smooth consumption over time. In the canonical framework, households accumulate during working life and then gradually decumulate in retirement while maintaining living standards (Ando & Modigliani, 1963). In practice, many households appear to reduce spending at retirement and draw down wealth more slowly than these models would imply. This combination, both retirees spending less and drawing down wealth more slowly, is puzzling because it suggests a more complex set of dynamics than traditional models of consumption smoothing would predict.

To what extent this phenomenon reflects a true decline in welfare, as opposed to changes in spending patterns or measurement issues, is an open question. Levels of consumption are often used as a proxy for welfare, but they may not fully capture the well-being of retirees, especially when higher consumption is compelled by medical needs or lower consumption is voluntary due to lifestyle changes. For example, Barrett and Brzozowski (2010) find that involuntary reductions in spending play a role. They show that households that retire unexpectedly, especially after adverse health shocks, can experience sharper spending adjustments than households that retire as planned. This suggests that some but not all of the observed decline in expenditure at retirement may be due to financial constraints rather than voluntary changes in spending patterns. This helps explain why average effects can mask substantial cross-household dispersion in retirement outcomes. However, some of the observed decline in expenditure may be due to retirees voluntarily reducing their spending because they become less active and therefore have less need for goods and services. For example, retirees may spend less on commuting, work clothing, and purchased meals as they transition out of the workforce (Aguiar & Hurst, 2008; Battistin et al., 2009). Additionally, if retirees substitute market purchases with home production, observed expenditure may fall while underlying consumption remains closer to smooth-path predictions (Becker, 1965; Hurd & Rohwedder, 2003).

Cross-country comparisons reinforce the importance of institutional context. Banks et al. (2016) show that spending profiles at older ages differ materially between the U.S. and U.K., with retirees in the U.S. having far higher consumption in late life compared to their U.K. counterparts. Retirees in the U.K. reduced their real consumption by 2.2% per year compared to 1.4% per year in the U.S. The researchers find that differences in healthcare financing and pension design are key drivers of these divergent patterns, with U.S. retirees facing

higher out-of-pocket healthcare costs, which thus drives higher spending in this category, especially at older ages when healthcare needs typically increase. In contrast, U.K. retirees have more comprehensive public healthcare coverage, which reduces out-of-pocket costs and therefore leads to lower healthcare spending in retirement. This results in a more pronounced decline in overall expenditure for U.K. retirees compared to U.S. retirees. However, when healthcare spending is excluded, the decline in expenditure at retirement is more similar between the two countries.

Declines in spending have also been observed in Australia. A review by the Grattan Institute (2019) finds that real spending declines by between 0.5% and 1% per year after retirement. Unlike the HES data for New Zealand, the Grattan Institute review examines longitudinal data, which allows them to track the same individuals over time. They find that the decline in spending is consistent between cross-sectional and longitudinal data, which suggests that the observed decline in spending with age is not solely due to differences between cohorts of retirees, but also reflects underlying changes in spending patterns as individuals age and is reflective of a less consumptive lifestyle.

Additionally, the Grattan Institute review finds that spending in Australia does not typically reduce in retirement due to financial constraints. Less than half of retirees in their study were drawing down wealth at all, and more than 40% were actually increasing their wealth during retirement. Retirees were also generally found to feel more financially comfortable than working-age people. As noted by Retirement Income Interest Group (2024), this suggests that the observed decline in spending at retirement is not primarily due to financial constraints, but instead reflects a combination of voluntary spending reductions and evolving lifestyle needs with age.

3.2 Bequests and Precautionary Savings

An aspect of the “retirement consumption puzzle” that has also been studied is why many retirees do not draw down significantly on their wealth during retirement, and in some cases even accumulate more wealth during retirement. This is a well-established phenomenon, having been observed in multiple countries and datasets. For example, King (1982) observe it in Canada, and the aforementioned Grattan Institute review finds it in Australia, and it has been extensively studied within the U.S. context (e.g., (De Nardi et al., 2010; Hildebrand, 2001; Love et al., 2009)).

An early explanation for this phenomenon (proposed by Hurd (1987)) is that the observed lack of decumulation is due to retirees’ desire to leave bequests to their heirs. In this view, retirees are not drawing down on their wealth because they want to preserve it for children or other beneficiaries. However, survey evidence and later studies have cast doubt on this explanation and proposed alternatives. A survey of Australian and Dutch retirees by Alonso-García et al. (2022) finds that intentional bequest motives are not a major concern for most respondents, aside from provision for a surviving spouse. Instead, retirees most often cite financial independence and quality of life. They conclude that limited decumulation may reflect precautionary saving for uncertain health and long-term care costs.

This “precautionary” saving motive is well supported in the literature. Modelling risk from uncertain health and long-term care expenditures, De Nardi et al. (2010) find that exposure to large, unexpected costs explains a substantial share of observed non-decumulation in the United States. As noted above, Banks et al. (2016) also show that healthcare costs are a major spending driver for U.S. retirees relative to U.K. retirees with broader public coverage. As Retirement Income Interest Group (2024) note, New Zealand retirees face a mixed system: most core medical services are publicly funded, but some services, particularly long-term care (for example assisted living and memory care), may still involve significant out-of-pocket costs (New Zealand Government, n.d.). This suggests that precautionary motives in New Zealand may be weaker than in the U.S., but likely remain meaningful for many households.

3.3 Spending by Household Composition and Wealth

Whether a retiree lives alone or with a partner is an important determinant of spending patterns, as is home-ownership status. We review this evidence in the next draft iteration once the remaining New Zealand-specific references for this subsection are finalised.

4 Modelling Perceptions of Risk and Utility

Traditional portfolio optimisation (arguably beginning with Markowitz (1952)) typically treats return variance as the risk measure. This follows from the assumption that investors are risk-averse and prefer lower variance

for a given expected return. However, for lifetime allocation problems, variance can be difficult to interpret and communicate. In particular, return variance does not directly capture the event investors often care most about: failing to meet spending needs in retirement. Rather, risk of ruin emerges from the interaction between return volatility and withdrawal requirements. For these reasons, variance-based optimisation may yield recommendations that are difficult to interpret and potentially misaligned with investor priorities.

This approach persists in the recent literature. For example, Mäkinen and Toivanen (2024) demonstrate their method using variance and semi-variance of terminal wealth as optimisation objectives for a control-matrix policy. While likely intended as a proof of concept, this objective choice can generate counterintuitive outcomes: the optimiser may truncate the upper tail of terminal wealth in ways that appear misaligned with practical investor concerns. A detailed discussion and replication are provided in the appendix.

Ultimately, whatever risk measure we choose should make sense from a behavioural perspective and, ideally, should be understandable to investors. As discussed in the preceding section, risk perceptions are closely tied to spending needs and concerns about meeting those needs over time. We therefore use that evidence to motivate a risk measure that is interpretable, communicable, and relevant to typical New Zealand retirees.

An essential component of optimising asset allocation strategies is the choice of risk measure, which serves as the objective function to be minimised (or maximised). This choice can materially influence the resulting strategy, because different measures capture different dimensions of risk and therefore imply different return-risk trade-offs.

Our risk measure should be interpretable and directly relevant to outcomes that real-world investors care about. Depending on the optimisation method, it may also need to be differentiable with respect to policy parameters, so that gradient-based methods can be used efficiently.

5 Methods for Simulating Lifetime Wealth Trajectories

A key component of studying lifetime asset allocation is the ability to simulate realistic wealth trajectories for individual investors. This requires modelling both accumulation and decumulation, while accounting for contributions, withdrawals, returns, and spending needs. Our approach uses a Monte Carlo framework that generates a wide range of possible outcomes under alternative allocation policies and market conditions. Importantly, the simulation framework is agnostic to return source: we can use historical, bootstrapped, or model-simulated returns. This flexibility allows us to test strategy robustness across a broad set of assumptions. The framework also accommodates time-varying deposits and withdrawals, which is essential for representing real-world household behaviour.

Unlike the approach proposed by Merton (1969), ours is a discrete-time simulation that models the investor's wealth at regular intervals. This allows us to capture savings and spending behaviour in fixed-dollar amounts, which is more realistic for most retirees.

Each step of the model takes the following form:

$$W_{t+1} = g_{t+1}W_t + \pi_{t+1} \quad (1)$$

Where W_t is the wealth at time t , g_{t+1} is the growth factor on the portfolio between time t and $t + 1$, and π_{t+1} is the net contribution (or withdrawal if negative) made at time $t + 1$. This can represent regular contributions during the accumulation phase or withdrawals to cover spending needs during the decumulation phase. The model can be easily adapted to include varying contribution and withdrawal amounts over time, allowing for a more nuanced representation of investor behaviour, including simulating variable or stochastic spending patterns.

The growth factor g_{t+1} is determined by the asset allocation strategy being tested, which specifies the proportion of wealth invested in different asset classes. Specifically, we define asset allocation as a vector $\mathbf{a}_t$ at time t , where each element $a_{i,t}$ represents the proportion of wealth invested in asset class i . The portfolio growth factor is then calculated as:

$$g_{t+1} = 1 + \mathbf{a}_t^\top \mathbf{r}_{t+1} \quad (2)$$

Where $\mathbf{r}_{t+1}$ is a vector of returns for each asset class between time t and $t + 1$.

In a Monte Carlo simulation, we generate a large number of possible return paths. This means that for each time step, we have N different return vectors for our k asset classes. For each path, we simulate the investor's wealth trajectory over time using the equations above. This means that equation 1 has the matrix form:

$$\mathbf{w}_{t+1} = \mathbf{w}_t \odot \mathbf{g}_{t+1} + \boldsymbol{\pi}_{t+1} \quad (3)$$

Where $\mathbf{w}_t$, $\mathbf{g}_{t+1}$, and $\boldsymbol{\pi}_{t+1}$ are vectors of length N representing the wealth, portfolio returns, and contributions or withdrawals for each of the N simulated paths at time t , and $\odot$ denotes element-wise (Hadamard) multiplication.

The growth factors $\mathbf{g}_{t+1}$ are determined by the asset allocation strategy and a $N \times k$ matrix of returns $\mathbf{R}_{t+1}$ for the k asset classes across the N simulated paths. The asset allocation strategy can be defined in various ways, the first of which is as a function of time (or age). In this case, we define the asset allocation at time t as $\mathbf{a}_t = f(t; \boldsymbol{\theta})$, where f is a function parameterised by a set of parameters $\boldsymbol{\theta}$.

This yields the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot f(t+1; \boldsymbol{\theta}) \quad (4)$$

In this form, the model can accommodate a wide range of asset allocation strategies based on the investor's age. This is similar to the approach that Anarkulova et al. (2025) use, and our first objective will be to replicate their findings for the optimal glide-paths, and to examine how these findings change when we modify the risk measures and spending assumptions to better reflect real-world investor behaviour, particularly in the New Zealand context.

Implications for Consilium and an Aside on Using Matrix Operations and Computational Efficiency

The effort taken to express the model in matrix form is not only for mathematical clarity; it also allows us to leverage vectorised operations through highly optimised linear algebra libraries (for example, `numpy` (Harris et al., 2020)), which can substantially accelerate simulation when path counts and horizon lengths are large. In prior internal work on related Monte Carlo workflows, non-vectorised implementations relying heavily on for-loops have been used, which are computationally inefficient. By contrast, matrix operations allow array-level computation and can produce order-of-magnitude speed improvements. Applied in practice, this may materially improve analysis throughput.

The use of matrix operations will also make it easier when it comes to optimising the asset allocation strategy, as we can leverage automatic differentiation tools (e.g. `tensorflow` (Martín Abadi et al., 2015), `pytorch` (Paszke et al., 2019)) to compute gradients of the risk measure with respect to the parameters of the asset allocation strategy.

5.1 Modelling Dynamic Investor Behaviour

The model described above is flexible enough to accommodate a wide range of asset allocation strategies, but it also allows for the modelling of dynamic investor behaviour in terms of contributions and withdrawals. This is an important feature, as it allows us to simulate more realistic spending patterns that can vary over time and in response to changes in the investor's wealth.

5.1.1 Wealth-Responsive Spending Policy

Rather than having a fixed spending pattern, we can also model the investor's spending as a function of their current wealth level. In some cases, retirees may have a fixed spending floor that they need to meet each year, regardless of the performance of their portfolio. In other cases, retirees may adjust their spending based on the performance of their portfolio, for example by reducing discretionary spending during periods of poor returns. To model this behaviour, we can define the net contribution (or withdrawal) at time t as a function of wealth: $\pi_t = g(W_t; \boldsymbol{\phi})$, where g is a function parameterised by a set of parameters $\boldsymbol{\phi}$. This allows us to simulate more realistic spending patterns that can vary over time and in response to changes in the investor's wealth. For example, we could model a spending policy that has a fixed floor and a variable discretionary component that adjusts based on the investor's current wealth level. This could be represented as:

$$\pi_t = -\max(S_{\text{floor}}, \alpha W_t) \quad (5)$$

Where S_{floor} is the fixed spending floor, and α is a parameter that determines how much of the investor's wealth is allocated to discretionary spending.

Additionally, we can prevent the investor from withdrawing more than their current wealth by defining the net contribution as:

$$\pi_t = -\min(g_{t+1} W_t, \max(S_{\text{floor}}, \alpha W_t)) \quad (6)$$

Equally, we could model a spending policy that has a random component, for example by adding a stochastic term to the spending function modelling rare, but significant, unexpected expenses that retirees may face, such as healthcare costs or home repairs.

5.1.2 Wealth-Responsive Allocation Policy

Like with the spending policy, we can also model the asset allocation strategy as a function of both time and wealth. In the age-based glide-path framework, the hypothetical investor’s asset allocation strategy is only responsive to time (i.e. age), and does not take into account the investor’s current wealth level. In practice, an investor’s risk tolerance and spending needs may vary significantly depending on their accumulated wealth at any given time. For example, an investor with a large portfolio may be able to tolerate more risk than an investor with a smaller portfolio, even if they are the same age. To address this limitation, we will attempt to extend the framework using a method similar to that proposed by Mäkinen and Toivanen (2024), who introduce a control matrix that allows the asset allocation strategy to depend on both time and wealth.

In this framework, the asset allocation at time t and wealth W_t is given by $\mathbf{a}_t = C(t, W_t; \boldsymbol{\theta})$, where C is the control matrix parameterised by $\boldsymbol{\theta}$. This yields the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot C(t + 1, W_t; \boldsymbol{\theta}) \quad (7)$$

The way that this control-matrix works is that it defines a grid over the time and wealth dimensions, with each cell in the grid corresponding to a specific asset allocation. During the simulation, the investor’s current age and wealth level are used to determine which cell in the grid they fall into, and thus the corresponding allocation to the risky asset (the remainder of the portfolio is allocated to the safe asset).

This approach allows for a more nuanced asset allocation strategy that can adapt to the investor’s changing circumstances over time. For example, an investor who has accumulated significant wealth may be able to take on more risk, while an investor with lower wealth may need to adopt a more conservative approach. However, it also exponentially increases the number of parameters that need to be optimised, which can make any optimisation process more challenging.

However, we can reduce the search-space somewhat by using less granular grid points for the control matrix and interpolating between them to determine the asset allocation for time steps and wealth levels that fall between the defined grid points. This allows us to maintain a reasonable level of flexibility in the asset allocation strategy while keeping the number of parameters manageable. Moreover, it is unlikely that investors’ asset allocation policy will be meaningfully different for small increments in time or wealth, so using a coarser grid with interpolation is unlikely to significantly reduce the optimality of the resulting strategy, while making the optimisation process more tractable. This idea of interpolating between grid points was first proposed by Mäkinen and Toivanen (2024), who use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined wealth-grid points, however, we also propose that interpolation could be used for time steps that fall between the defined time-grid points as well, which would further reduce the number of parameters that need to be optimised.

To illustrate this more concretely, consider Figure 2, which shows a simple control matrix for a two-asset portfolio. The matrix has six time steps (0 to 5) and five wealth levels (0 to 4), giving a 6 by 5 grid. Each cell corresponds to a risky-asset allocation (with the remainder allocated to the safe asset). For instance, if an investor is at time step 2 with wealth level 3, the policy returns the allocation in that cell. For time or wealth values between grid points, we use the interpolation method proposed by Mäkinen and Toivanen (2024). For example, at time 2.5 and wealth 3.5, bilinear interpolation uses the four neighbouring grid points to determine allocation.

For allocations with more than two asset classes, the control matrix can be extended to a higher-dimensional tensor, with each additional dimension representing an additional asset class. Mäkinen and Toivanen (2024) use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined grid points, ensuring a smooth transition between different allocation strategies.

5.2 Methods for Simulating Asset Returns

6 Methods for Optimising Asset Allocation Strategies

The key challenge in optimising an asset allocation strategy is that the parameter space can be extremely large, even for simple policy classes. For example, for a two-asset age-based glide-path with allocations in 10 percent

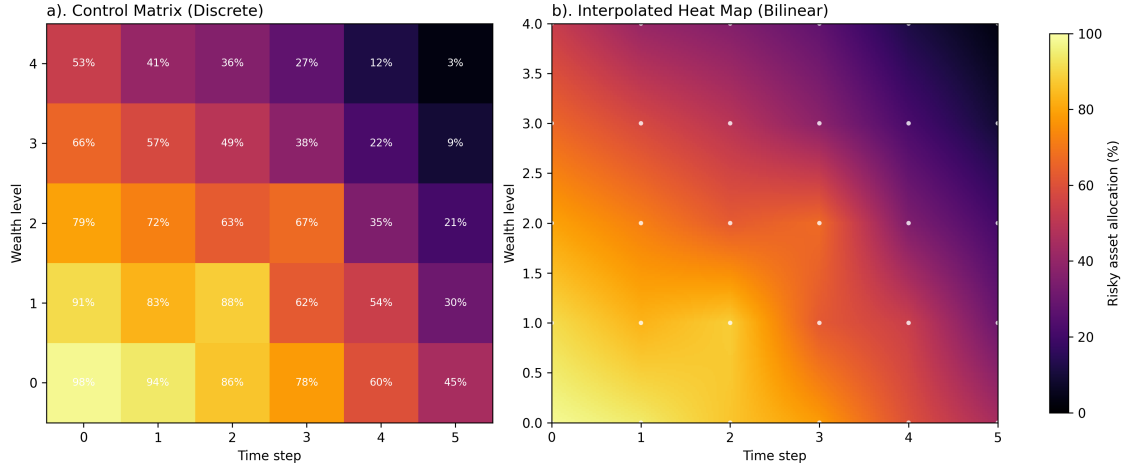


Figure 2: Illustrative example of a discrete control matrix (a) and bilinearly interpolated allocation surface (b).

increments over 60 years, there are 11 possible allocations at each time step, giving 11^{60} possible strategies. This search space is intractable, so numerical optimisation is required.

Merton (1969) use a formulation in which the investor maximises a lifetime utility function over consumption. The framework is mathematically tractable, but it relies on proportional consumption from wealth each period, which is often unrealistic for retirees facing fixed-dollar spending needs. In addition, the closed-form solution assumes IID returns, precisely the return structure that Anarkulova et al. (2025) argue can bias conclusions about declining-risk optimality. For our purposes, this framework is therefore limited.

Anarkulova et al. (2025) use a recursive approach in which grid search is used to update allocation at each time step, with forward passes repeated until convergence. This approach is likely suitable for age-based glide-path optimisation, but extension to a time-by-wealth control matrix is less straightforward because the parameter count increases by roughly an additional order of magnitude.

In their paper, Mäkinen and Toivanen (2024) optimise a control-matrix allocation strategy using gradient-based methods and an adjoint formulation, which is computationally appealing for high-dimensional policy spaces. This provides a practical starting point for our own optimisation design, while still allowing us to adapt the objective function to better reflect retiree-relevant risk.

7 Optimal Lifetime Asset Allocation Strategies for New Zealand Retirees

Having established the simulation and optimisation framework, we now apply it to evaluate candidate lifetime allocation policies in the New Zealand retirement context.

7.1 Optimal Age-Based Glide-Paths

7.2 Optimal Wealth-Responsive Allocation Strategies

8 Discussion and Implications

9 Conclusion

References

- Aguiar, M., & Hurst, E. (2008). *Deconstructing lifecycle expenditure* (tech. rep. No. 13893). National Bureau of Economic Research. <https://doi.org/10.3386/w13893>
- Alonso-García, J., Bateman, H., Bonekamp, J., van Soest, A., & Stevens, R. (2022). Saving preferences after retirement. *Journal of Economic Behavior & Organization*, 198, 409–433. <https://doi.org/10.1016/j.jebo.2022.04.005>
- Anarkulova, A., Cederburg, S., & O’Doherty, M. S. (2025, March). Beyond the status quo: A critical assessment of lifecycle investment advice [SSRN Working Paper]. <https://doi.org/10.2139/ssrn.4590406>
- Ando, A., & Modigliani, F. (1963). The “life cycle” hypothesis of saving: Aggregate implications and tests. *The American Economic Review*, 53(1), 55–84. <https://www.jstor.org/stable/1817129>

- Australian Taxation Office. (n.d.). *Super guarantee percentage* [Retrieved 6 July 2023].
- Banks, J., Blundell, R., Levell, P., & Smith, J. P. (2016). *Life-cycle consumption patterns at older ages in the us and the uk: Can medical expenditures explain the difference?* (Tech. rep. No. 22513). National Bureau of Economic Research. <https://doi.org/10.3386/w22513>
- Barrett, G., & Brzozowski, M. (2010). *Involuntary retirement and the resolution of the retirement-consumption puzzle: Evidence in australia* (tech. rep.). University of Sydney, School of Economics.
- Battistin, E., Brugiavini, A., Rettore, E., & Weber, G. (2009). The retirement consumption puzzle: Evidence from a regression discontinuity approach. *American Economic Review*, 99(5), 2209–2226. <https://doi.org/10.1257/aer.99.5.2209>
- Becker, G. S. (1965). A theory of the allocation of time. *The Economic Journal*, 75(299), 493–517. <https://doi.org/10.2307/2228949>
- Bengen, W. P. (1996). Asset allocation for a lifetime. *Journal of Financial Planning*, 9(4), 86–92. <https://www.financialplanningassociation.org/article/journal/AUG96-asset-allocation-lifetime>
- Bodie, Z., Kane, A., & Marcus, A. J. (2023). *Investments*. McGraw-Hill US Higher Ed ISE. <https://ebookcentral.proquest.com/lib/canterbury/detail.action?docID=7190049>
- Bodie, Z., Merton, R. C., & Samuelson, W. F. (1992). Labor supply flexibility and portfolio choice in a life cycle model. *Journal of Economic Dynamics and Control*, 16(3), 427–449. [https://doi.org/https://doi.org/10.1016/0165-1889\(92\)90044-F](https://doi.org/https://doi.org/10.1016/0165-1889(92)90044-F)
- Campbell, J. Y. (1996). Understanding risk and return. *Journal of Political Economy*, 104(2), 298–345.
- Choi, J. J. (2022). Popular personal financial advice versus the professors. *Journal of Economic Perspectives*, 36(4), 167–192. <https://www.jstor.org/stable/27171135>
- Cocco, J. F., Gomes, F. J., & Maenhout, P. J. (2005). Consumption and portfolio choice over the life cycle. *The Review of Financial Studies*, 18(2), 491–533. Retrieved December 8, 2025, from <http://www.jstor.org/stable/3598044>
- De Nardi, M., French, E., & Jones, J. B. (2010). *Differential mortality, uncertain medical expenses, and the saving of elderly singles* (tech. rep. No. 16453). National Bureau of Economic Research. <https://doi.org/10.3386/w16453>
- Dexter, G. (2025, November 24). *Kiwisaver provider calls for increased contribution to be compulsory*. Retrieved December 6, 2025, from <https://www.rnz.co.nz/news/political/579777/kiwisaver-provider-calls-for-increased-contribution-to-be-compulsory>
- Dolvin, S. D., Templeton, W. K., & Rieber, W. (2010). Asset allocation for retirement: Simple heuristics and target-date funds. *Scholarship and Professional Work - Business*, (17). https://digitalcommons.butler.edu/cob_papers/17
- Estrada, J. (2014). The glidepath illusion: An international perspective. *The Journal of Portfolio Management*, 40(4), 52–64. <https://doi.org/10.3905/jpm.2014.40.4.052>
- Estrada, J. (2015). The retirement glidepath: An international perspective. *The Journal of Investing*, 25(2), 28–36. <https://doi.org/10.3905/joi.2016.25.2.028>
- Financial Markets Authority. (2023, August 9). *Kiwisaver default funds*. Retrieved December 6, 2025, from <https://www.fma.govt.nz/consumer/kiwisaver-and-superannuation/about-kiwisaver/kiwisaver-default-funds/>
- Financial Markets Authority. (2024, June 24). *Spotlight on: Balanced funds*. Retrieved December 6, 2025, from <https://www.fma.govt.nz/library/articles/spotlight-on-balanced-funds/>
- Ghysels, E., Santa-Clara, P., & Valkanov, R. (2005). There is a risk-return trade-off after all. *Journal of Financial Economics*, 76(3), 509.
- Gilbert, A. (2025, November 27). *Lifting kiwisaver contributions to 12% makes sense – when the whole scheme is fixed*. Retrieved December 6, 2025, from <https://www.nzherald.co.nz/nz/lifting-kiwisaver-contributions-to-12-makes-sense-when-the-whole-scheme-is-fixed>
- Grattan Institute. (2019). *The savings behaviour of older households: Submission 435 – supplementary submission to inquiry into the implications of removing refundable franking credits* (Supplementary submission No. 435) (Draws from: Daley, J., Coates, B., Wiltshire, T., Emslie, O., Nolan, J., and Chen, T. (2018), “Money in Retirement: More than Enough”, Grattan Institute). Parliament of Australia.
- Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>

- Hildebrand, V. (2001). *Wealth accumulation of us households: What do we learn from the sipp data?* (Social and Economic Dimensions of an Aging Population Research Papers). McMaster University. <https://EconPapers.repec.org/RePEc:mcm:sedapp:41>
- Hurd, M. D. (1987). Savings of the elderly and desired bequests. *The American Economic Review*, 77(3), 298–312. <https://www.jstor.org/stable/1804093>
- Hurd, M. D., & Rohwedder, S. (2003). *The retirement-consumption puzzle: Anticipated and actual declines in spending at retirement* (tech. rep. No. 9586). National Bureau of Economic Research. <https://doi.org/10.3386/w9586>
- Ibbotson, R., & Kaplan, P. (2001). Does asset allocation policy explain 40, 90, 100 percent of performance? *Yale School of Management, Yale School of Management Working Papers*, 56. <https://doi.org/10.2469/faj.v56.n1.2327>
- King, M. A. (1982). Asset holdings and the life-cycle. *The Economic Journal*, 92(366), 247–267. <https://doi.org/10.2307/2232439>
- Le, T., & Richardson, E. (2023, August). *Expenditure patterns of new zealand retiree households* (Motu Working Paper No. 23-07). Motu Economic and Public Policy Research. <https://motu.nz/our-research/population-and-labour/individual-and-group-outcomes/expenditure-patterns-nz-retiree-households>
- Love, D. A., Palumbo, M. G., & Smith, P. A. (2009). The trajectory of wealth in retirement. *Journal of Public Economics*, 93(1), 191–208. <https://doi.org/https://doi.org/10.1016/j.jpubeco.2008.09.003>
- Lundblad, C. (2007). The risk return tradeoff in the long run: 1836–2003. *Journal of Financial Economics*, 85(1), 123–150. <https://doi.org/10.1016/j.jfineco.2006.06.003>
- Luxon, C. (2025, November 23). *National to lift kiwisaver contributions*. Retrieved December 6, 2025, from <https://www.national.org.nz/news/20251123-hon-christopher-luxon-national-to-lift-kiwisaver-contributions>
- Mäkinen, R. A. E., & Toivanen, J. (2024). Monte carlo expected wealth and risk measure trade-off portfolio optimization. *SIAM Journal on Financial Mathematics*.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91. Retrieved August 6, 2025, from <http://www.jstor.org/stable/2975974>
- Martín Abadi, Ashish Agarwal, Paul Barham, Eugene Brevdo, Zhifeng Chen, Craig Citro, Greg S. Corrado, Andy Davis, Jeffrey Dean, Matthieu Devin, Sanjay Ghemawat, Ian Goodfellow, Andrew Harp, Geoffrey Irving, Michael Isard, Jia, Y., Rafal Jozefowicz, Lukasz Kaiser, Manjunath Kudlur, . . . Xiaoqiang Zheng. (2015). TensorFlow: Large-scale machine learning on heterogeneous systems [Software available from [tensorflow.org](https://www.tensorflow.org/)]. <https://www.tensorflow.org/>
- Merton, R. C. (1969). Lifetime portfolio selection under uncertainty: The continuous-time case. *The Review of Economics and Statistics*, 51(3), 247–257. <https://doi.org/10.2307/1926560>
- Ministry of Business, Innovation and Employment and The Treasury. (2019, August 7). *Review of kiwisaver default provider arrangements* [Consultation document]. Retrieved December 6, 2025, from <https://www.treasury.govt.nz/publications/consultation/review-kiwisaver-default-provider-arrangements>
- Ministry of Social Development. (2022). *Description of new zealand’s current retirement income policies* (tech. rep.) (Background paper prepared for the Retirement Commissioner’s 2022 Review of Retirement Income Policy). Ministry of Social Development. Retrieved March 17, 2026, from <https://assets.retirement.govt.nz/public/Uploads/Retirement-Income-Policy-Review/2022-RRIP/Description-of-New-Zealands-Current-Retirement-Income-Policies-MSD-for-2022-RRIP-.pdf>
- New Zealand Government. (n.d.). *Paying for residential care*. Retrieved April 13, 2026, from <https://www.govt.nz/browse/health/rest-homes-and-residential-care/pay-for-residential-care/>
- OECD. (2025). *Pension markets in focus 2025*. OECD Publishing. Paris. <https://doi.org/10.1787/b095d0a0-en>
- Palmer, R. (2025, September 7). *Watch: Peters promises compulsory 10% kiwisaver and tax cuts*. Retrieved December 6, 2025, from <https://www.rnz.co.nz/news/national/572352/watch-peters-promises-compulsory-10-percent-kiwisaver-and-tax-cuts>
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., . . . Chintala, S. (2019). Pytorch: An imperative style, high-performance deep learning library. <https://arxiv.org/abs/1912.01703>
- Retirement Commission Te Ara Ahunga Ora. (n.d.). *A snapshot of the history of kiwisaver* [PDF pamphlet]. Retrieved December 6, 2025, from <https://assets.retirement.govt.nz/public/Uploads/Retirement-Income-Policy-Review/TAAO-A-Snapshot-of-the-History-of-KiwiSaver.pdf>

- Retirement Commission Te Ara Ahunga Ora. (2022, August 15). *New research highlights how reliant older new zealanders are on nz super*. Retrieved March 17, 2026, from <https://retirement.govt.nz/news/latest-news/new-research-highlights-how-reliant-older-new-zealanders-are-on-nz-super>
- Retirement Income Interest Group. (2024, December). *Spending patterns through retirement: Implications for retirement planning and drawdown* (Report). New Zealand Society of Actuaries. <https://actuaries.org.nz/content/uploads/2024/12/Spending-through-retirement-RIIG-Dec2024.pdf>
- Shiller, R. (2005). Life-cycle portfolios as government policy. *The Economists' Voice*, 2(1), 14–14. <https://doi.org/10.2202/1553-3832.1095>
- Shiller, R. J. (2006). Life-cycle personal accounts proposal for social security: An evaluation of president bush's proposal. *Journal of Policy Modeling*, 28(4), 427–444. <https://doi.org/https://doi.org/10.1016/j.jpolmod.2005.10.010>
- SuperLife. (2026, March 26). *Statement of investment policy and objectives (sipo): Superlife invest*. Retrieved March 31, 2026, from https://www.superlife.co.nz/files/LegalDocs/SIPO/SuperLifeInvest-Statement_of_Investment_Policy_and_Objectives.pdf
- Swoboda, K. (2014, March 11). *Major superannuation and retirement income changes in australia: A chronology* (Research Papers 2013–14). Australian Parliamentary Library.
- Te Ara Ahunga Ora: Retirement Commission. (2021). *Review of retirement income: Policy paper 03* (Policy Paper No. 03). Te Ara Ahunga Ora: Retirement Commission. <https://assets.retirement.govt.nz/public/Uploads/Retirement-Income-Policy-Review/TAAO-RC-Policy-Paper-2021-03.pdf>
- The Treasury. (2025, September 24). *He tirohanga mokopuna 2025: Statement on the long term fiscal position* (Retrieved from <https://www.treasury.govt.nz/publications/lftp/he-tirohanga-mokopuna-2025> on December 6, 2025). The Treasury. New Zealand Government. Retrieved December 6, 2025, from <https://www.treasury.govt.nz/publications/lftp/he-tirohanga-mokopuna-2025>

A Replication and Discussion of Control Matrix Optimisation as Proposed by Mäkinen and Toivanen (2024)

Mäkinen and Toivanen (2024) formulate their Monte Carlo optimisation in a way that is similar to the framework described in the main text. As in our setup, they use a control matrix that allows asset allocation to respond jointly to age and wealth. Their objective function is based on terminal wealth moments, specifically variance or semi-variance. They then compute gradients of this objective with respect to control-matrix parameters using adjoint methods, which enables efficient gradient-based optimisation (in their case, a quasi-Newton method with BFGS updates).

Because this method is mathematically elegant and computationally efficient, we have attempted to replicate it here. The code for this replication can be found in the accompanying repository, and the results are shown in Figure 3 for the mean-variance and mean-semi-variance cases.

In line with the setup described by Mäkinen and Toivanen (2024), the replication uses a Monte Carlo horizon of $T = 20$ years with time step $\Delta t = 0.25$, giving $N = 80$ time steps. The policy is represented on a time-wealth control grid with $M = 31$ wealth nodes, upper wealth bound $W_{\max} = 30$, and bilinear interpolation between nodes. In the notebook, wealth is clipped to $[0, W_{\max}]$ before interpolation. Optimisation is performed with L-BFGS-B (a quasi-Newton method) with box constraints.

For the one-risky-asset, no-leverage case used in Figure 3, the full parameterisation is summarised in Table 1.

Table 1: Parameter setup for the one-risky-asset mean-variance replication.

Parameter	Value
Risk-free rate (r)	0.03
Risky-asset drift (μ_1)	0.0795
Risky-asset volatility (σ_1)	0.15
Contribution rate (π)	0.1
Initial wealth (w_0)	1
Minimum risky allocation ($p_{\min}$)	0
Maximum risky allocation ($p_{\max}$)	1
Number of simulation paths (K)	10^5
Risk-aversion weight (λ)	0.1

The replication results are broadly consistent with those reported by Mäkinen and Toivanen (2024). However, they also highlight important limitations. First, as an objective for a retirement investor, terminal-wealth variance is not especially intuitive. In our replication, the terminal-wealth distribution becomes bimodal because the optimiser shifts to a highly conservative policy at high wealth levels, truncating the upper tail that would normally appear under a log-normal-like process. Although this behaviour is mathematically consistent with variance minimisation, it is not obviously aligned with investor welfare. A similar pattern appears under semi-variance, albeit at higher wealth levels. Because semi-variance is computed relative to the mean, higher-return strategies can increase the penalty by raising the mean itself, again encouraging upper-tail truncation. This reinforces the need for risk measures that are both interpretable and behaviourally relevant.

Additionally, the control matrix has limited effective resolution in early periods: many simulated paths map to only a few low-wealth nodes because initial wealth is far below the fixed upper wealth bound. This reduces the optimiser’s ability to distinguish trajectories precisely when policy differences likely matter most due to compounding. It also creates many redundant parameters at wealth levels that are never reached (upper-left region in Figure 3). For this reason, we propose calibrating the control-matrix upper bound to the simulated wealth distribution so that resolution is concentrated where paths actually lie.

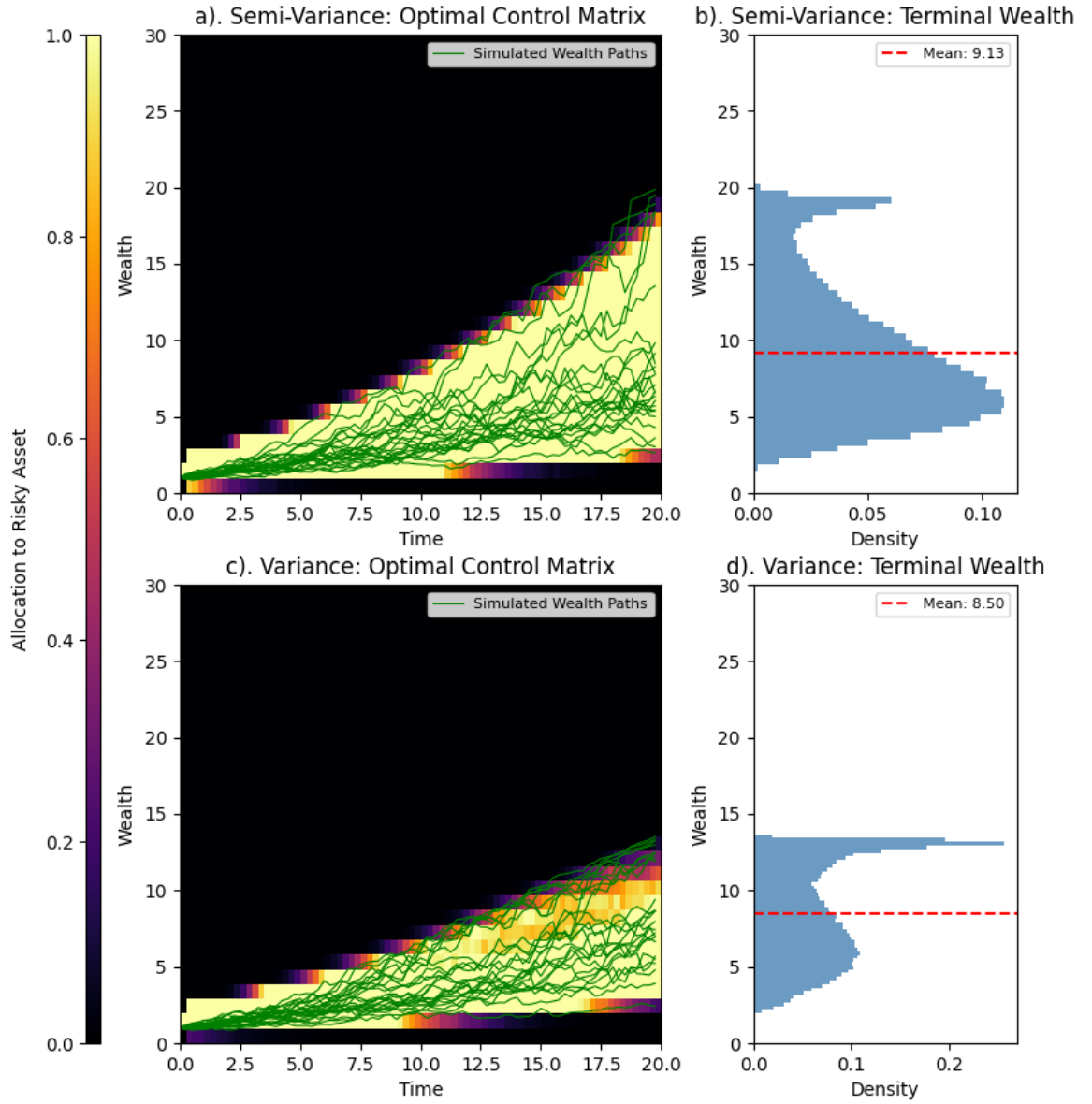


Figure 3: Replication of the control matrix optimization method proposed by Mäkinen and Toivanen (2024) for one risky asset and no leverage. The left column shows the optimal control matrix for the mean-semi-variance case (top) and the mean-variance case (bottom). The right column shows the corresponding histograms of terminal wealth for each risk measure respectively.